Unit 1: Internet of Things (EIoT)

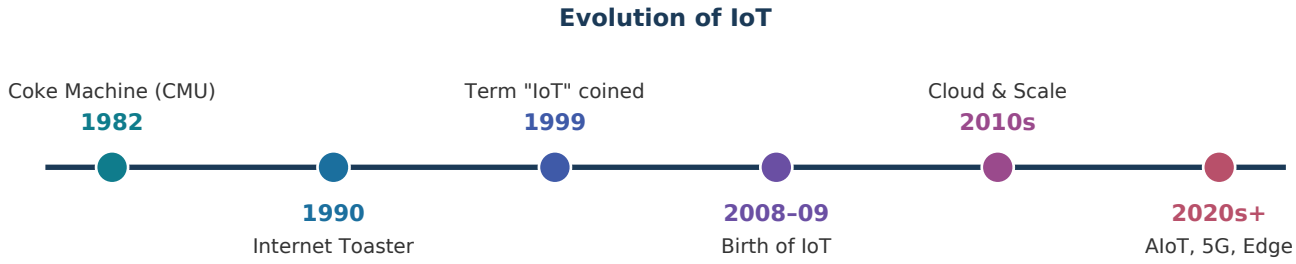
Question Bank & Answer Key

Section A: 10-Mark Questions

Q1. Define the Internet of Things (IoT) and trace its Genesis and evolution from 1982 to the 2020s+ era. Explain its foundational concept and key milestone transitions.

Answer:

The **Internet of Things (IoT)** is a network of interconnected physical objects ("things") embedded with sensors, software, and communication technologies that enable them to collect, exchange, and act on data over the internet without human intervention.

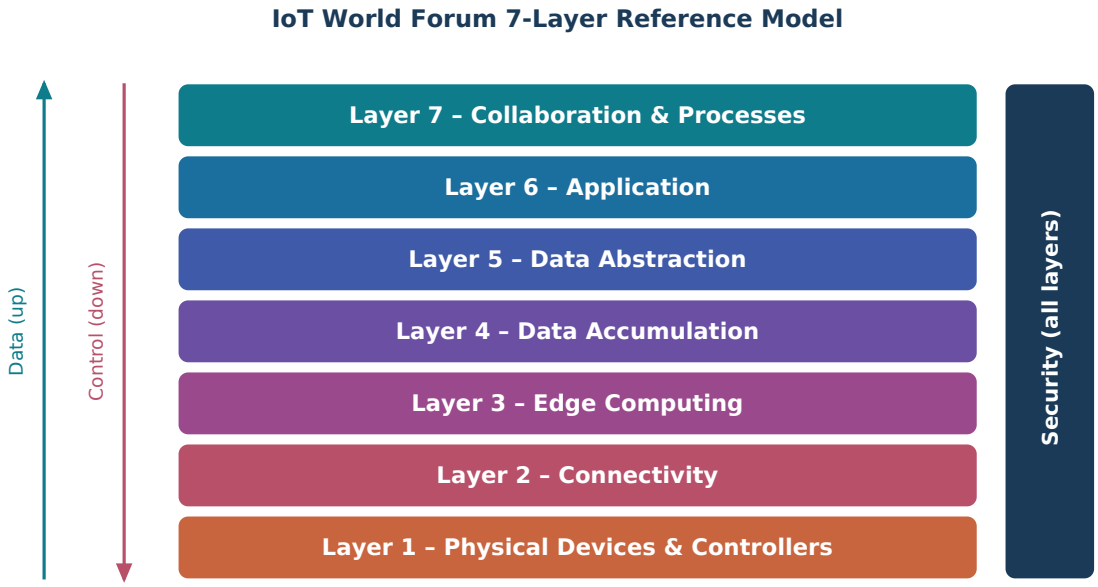


1. **Definition and Paradigm Shift:** IoT moves communication from human-to-human interaction to autonomous **machine-to-machine (M2M)** data exchange. Physical devices collect environmental parameters and execute actions automatically via cloud or edge processing.
2. **1982 – First IoT-like Device:** The genesis of IoT began at Carnegie Mellon University with a connected **Coke machine**. The machine reported its inventory and the temperature of cold drinks remotely over an early network connection.
3. **1990 – Early Internet Automation:** John Romkey developed the first internet-controlled appliance, a **toaster** that could be powered on and off over the Internet. This demonstrated basic remote actuation capabilities over IP networks.
4. **1999 – Coining the Term "IoT":** Kevin Ashton officially coined the term **"Internet of Things"** while working at the MIT Auto-ID Center. He envisioned using RFID tags to link physical assets to the internet for supply chain tracking.
5. **2008–2009 – The Birth of IoT:** This period marked the official birth of IoT, defined as the exact point in history when **more physical "things" were connected to the internet than human beings** on the planet.
6. **2010s – Scalability and Cloud Integration:** Cheap silicon sensors, widespread Wi-Fi connectivity, smartphones, and **cloud computing platforms** enabled massive consumer and industrial IoT scaling. Devices gained centralized data processing and remote mobile control.
7. **2020s+ – Era of AIoT, 5G, and Edge:** Modern IoT integrates **Artificial Intelligence (AIoT)**, 5G cellular connectivity, and edge computing. This enables real-time analytics, ultra-low latency control, and autonomous operation across critical systems.

8. **Core Drivers of Genesis:** Evolution was driven by hardware miniaturization, falling sensor costs, pervasive wireless coverage, and the expansion of IP address space via **IPv6**.
9. **Economic Impact Shift:** IoT evolved from isolated novelty projects into a major global driver of **industrial productivity**, predictive maintenance, and new service-based business models.
10. **Modern Technological Convergence:** Today's IoT represents the full convergence of physical operational technology (OT) with enterprise information technology (IT), enabling digital twin modeling and smart automation.

Q2. Detail the IoT World Forum 7-Layer Reference Model. Explain the specific role, core functions, and real-world examples for each layer from physical devices to collaboration processes.

Answer:



1. **Layer 1 – Physical Devices & Controllers:** This foundational layer contains physical objects ("things") embedded with sensors, actuators, and microcontrollers. It converts physical parameters (temperature, motion, light) into digital data and executes physical actions. *Example: Industrial vibration sensors and solenoid valves.*
2. **Layer 2 – Connectivity:** Focuses on establishing reliable, secure communication channels between physical devices and processing hubs. It encompasses physical link protocols (Wi-Fi, Ethernet, 802.15.4) and routing protocols. *Example: A cellular 5G gateway forwarding sensor frames.*
3. **Layer 3 – Edge Computing:** Performs immediate data filtering, normalization, and local processing close to the data source. It reduces latency and bandwidth demands by converting raw sensor streams into actionable events. *Example: An edge gateway detecting machine anomalies and shutting down equipment locally.*
4. **Layer 4 – Data Accumulation:** Serves as the storage staging area where stream data is captured and stored in databases. It converts dynamic in-flight network packets into static stored data for historical analysis. *Example: InfluxDB time-series storage on a cloud server.*
5. **Layer 5 – Data Abstraction:** Aggregates and structures stored data from heterogeneous sources, making it uniform for application consumption. It manages data access via APIs and decouples raw storage from applications. *Example: Exposing structured JSON payloads via RESTful APIs.*
6. **Layer 6 – Application:** Delivers user-facing software applications, performance dashboards, reporting tools, and control interfaces. It executes domain-specific logic for smart homes,

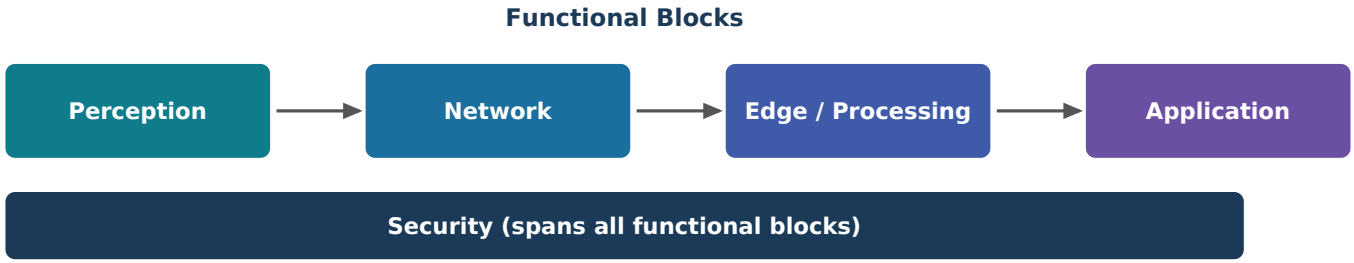
healthcare, or smart grids. *Example: A smartphone mobile app showing room humidity and AC status.*

7. **Layer 7 – Collaboration & Processes:** Integrates IoT application data into enterprise workflows, business logic, and human decision-making processes. It links technical metrics directly to economic outcomes and organizational goals. *Example: Predictive maintenance alerts automatically triggering ERP supply-chain order tickets.*
8. **Vertical Security Architecture:** Security mechanisms (encryption, authentication, authorization) span across all 7 layers to ensure end-to-end data integrity.
9. **Data Flow Directionality:** Data moves upward from Layer 1 to Layer 7 during sensing and reporting, while control commands move downward from Layer 7 to Layer 1 for physical actuation.
10. **Scalability and Standardization:** The 7-layer model simplifies IoT system design by allowing vendors to build interoperable products at specific functional layers.

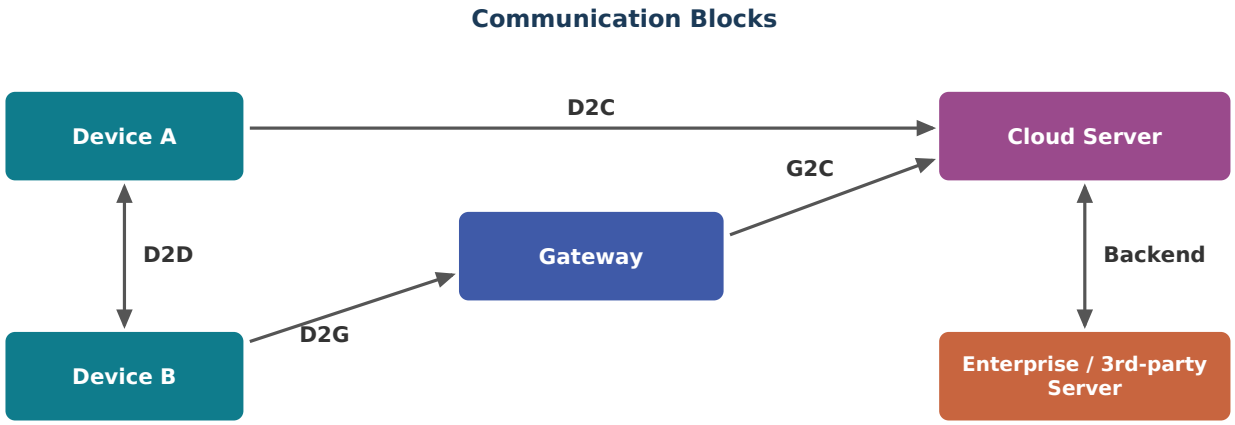
Q3. Explain the Logical Design of IoT. Detail its core functional blocks (Perception, Network, Edge/Processing, Application, Security) and its five primary communication blocks (D2D, D2G, D2C, G2C, Backend).

Answer:

The **Logical Design of IoT** is an abstract representation of system functionality and data flow, defining functional capability blocks and interaction models without detailing physical hardware implementations.



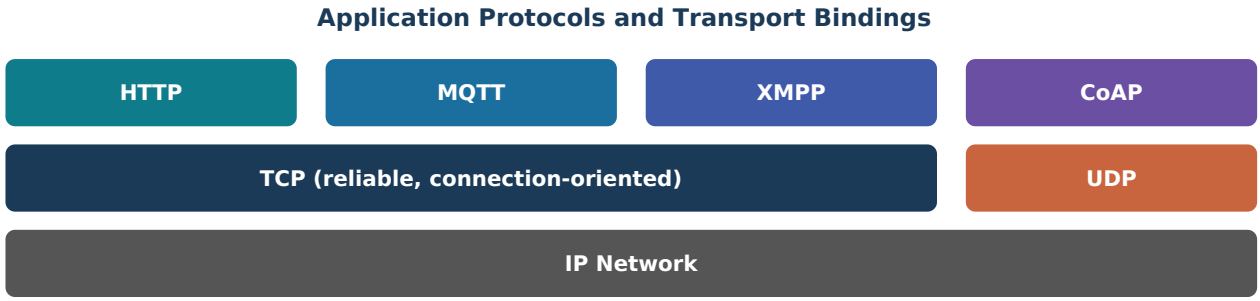
1. **Perception Functional Block:** Manages physical environment interaction using hardware sensors, actuators, and RFID devices. It converts environmental states into digital readings and executes physical movements.
2. **Network Functional Block:** Responsible for secure, reliable data transport across communication infrastructures. It utilizes wireless/wired standards like Zigbee, Wi-Fi, 6LoWPAN, and 5G.
3. **Edge/Processing Functional Block:** Handles data filtering, local computation, storage, and analytics. It processes data on local fog nodes for low latency or routes it to cloud infrastructure for heavy compute.
4. **Application Functional Block:** Intercepts processed data to provide domain-specific services and user control interfaces. It drives smart lighting, health monitoring, and industrial dashboards.
5. **Security Functional Block:** Operates cross-functionally across all blocks to provide device identity, message encryption, access control, and threat mitigation.



6. **Device-to-Device (D2D) Communication:** Direct short-range communication between devices without intermediate infrastructure. *Example: A smart light switch talking directly to a smart bulb via Bluetooth.*
7. **Device-to-Gateway (D2G) Communication:** Devices connect to an intermediate gateway that performs protocol conversion, data filtering, and local routing. *Example: A wearable pulse sensor connecting to a smartphone.*
8. **Device-to-Cloud (D2C) Communication:** IoT devices connect directly to cloud servers over IP networks using protocols like MQTT or HTTP. *Example: A Wi-Fi security camera streaming directly to AWS IoT Core.*
9. **Gateway-to-Cloud (G2C) Communication:** Gateways aggregate data from dozens of local low-power sensors and forward consolidated packets to cloud platforms. *Example: A smart home hub sending sensor data over 4G to cloud storage.*
10. **Backend (Server-to-Server) Communication:** Data exchange between cloud servers, enterprise databases, and third-party analytics platforms using RESTful APIs.

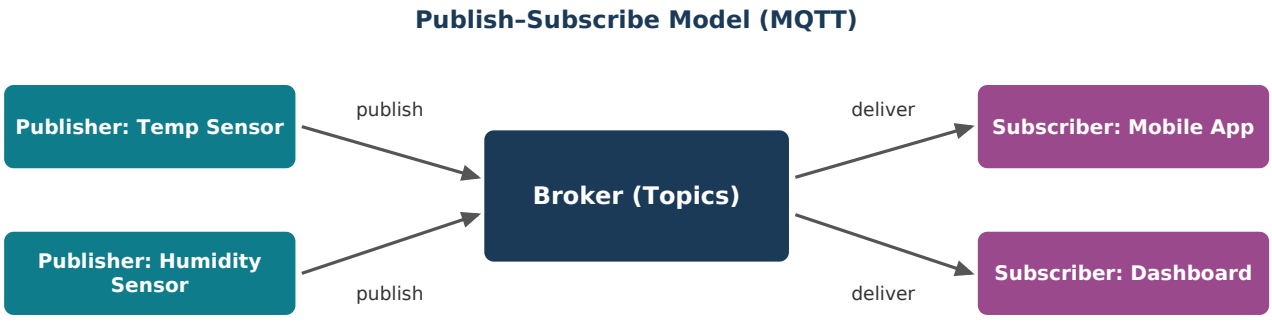
Q4. Provide a comprehensive architectural analysis of Application Layer Protocols in IoT: HTTP, CoAP, MQTT, and XMPP. Detail their messaging models, transport bindings, overhead, and ideal deployment scenarios.

Answer:



1. **HTTP (Hypertext Transfer Protocol):** A heavy, text-based, request-response protocol running over TCP (Port 80/443). It uses verb methods (GET, POST, PUT, DELETE) and is stateless.
2. **HTTP Limitations in IoT:** High header overhead (hundreds of bytes) and TCP connection setup make HTTP inefficient for resource-constrained, battery-operated IoT sensors.

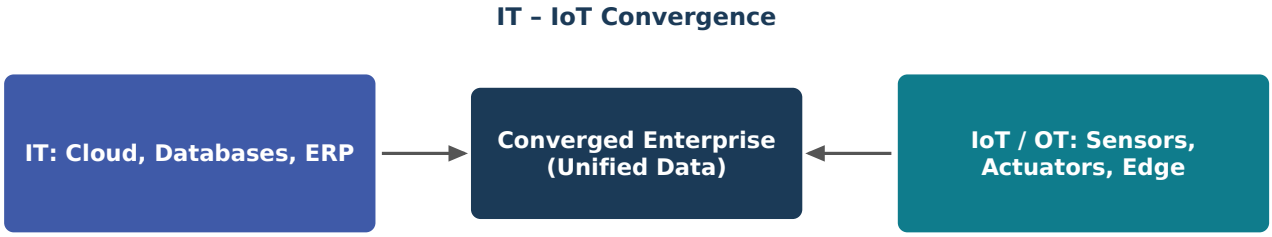
3. **CoAP (Constrained Application Protocol):** A lightweight binary protocol specifically designed for constrained low-power IoT devices operating on lossy networks.
4. **CoAP Architecture & Features:** Runs over UDP using a 4-byte header, adopting a RESTful model (GET, POST) with asynchronous confirmable (CON) and non-confirmable (NON) messages.
5. **MQTT (Message Queuing Telemetry Transport):** A lightweight, publish-subscribe messaging protocol running over TCP, optimized for constrained networks and remote monitoring.
6. **MQTT Broker Architecture:** Clients connect to a central broker, publishing messages to specific text topics or subscribing to topics to receive automated updates.



7. **MQTT Quality of Service (QoS):** Offers three QoS levels: QoS 0 (At most once / fire and forget), QoS 1 (At least once / acknowledged), and QoS 2 (Exactly once / four-way handshake).
8. **XMPP (Extensible Messaging and Presence Protocol):** An open, XML-based protocol designed for instant messaging, presence verification, and real-time state synchronization.
9. **XMPP Usage in IoT:** Uses persistent XML streams over TCP, enabling structured addressing (JIDs) for direct device-to-device addressing, though XML parsing adds processing overhead.
10. **Protocol Selection Trade-offs:** CoAP (UDP) is best for constrained battery devices; MQTT (TCP) is ideal for reliable event publish-subscribe; HTTP is used for cloud APIs; XMPP handles structured messaging.

Q5. Discuss the Convergence of IT and IoT alongside the primary Drivers Behind New Network Architectures in modern enterprise digital transformation.

Answer:



1. **Definition of IT-IoT Convergence:** Merges traditional IT (cloud, databases, ERPs) with IoT operational physical devices (sensors, actuators, real-time edge networks).
2. **Unified Data Flow:** Real-time physical operational data streams from IoT devices directly into enterprise IT data lakes for predictive analytics and corporate decision-making.
3. **Enhanced Operational Efficiency:** Convergence enables automated workflows, such as IoT machinery sensors automatically ordering replacement parts in an IT ERP system.
4. **Hybrid Infrastructure Shift:** Blends centralized IT cloud computing with distributed IoT edge gateways to balance massive computation against real-time operational needs.
5. **Expanded Security Attack Surface:** Merging physical devices into IT networks introduces vulnerabilities, requiring unified zero-trust security policies.
6. **Driver 1 – Massive Connected Device Growth:** Billions of M2M devices entering networks overwhelm traditional human-centric client-server architectures, demanding scalable network structures.
7. **Driver 2 – Ultra-Low Latency Demands:** Mission-critical applications (autonomous vehicles, industrial robotics) require sub-millisecond response times, forcing compute to the network edge.
8. **Driver 3 – Massive Data Volumes:** High-bandwidth video and telemetry streams create cloud bottlenecks, driving edge filtering and localized processing architectures.
9. **Driver 4 – Network Virtualization (SDN/NFV):** Software-Defined Networking and Network Function Virtualization enable dynamic network slicing tailored to specific IoT traffic classes.
10. **Driver 5 – Power and Cost Constraints:** Low-Power Wide-Area Networks (LPWANs) and energy-efficient protocols are required to support battery-operated remote devices over long lifespans.

Section B: 5-Mark Questions

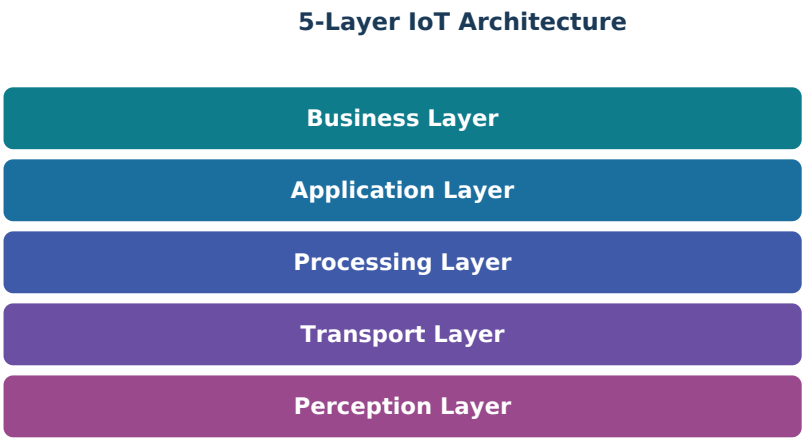
Q6. Differentiate between the Traditional Internet and the Internet of Things (IoT) across focus, primary devices, communication modes, data types, and application examples.

Answer:

Comparative Parameter	Traditional Internet	Internet of Things (IoT)
Primary Focus	Human-centric communication, content access, and web browsing.	Machine-to-machine (M2M) communication and automated sensing.
Connected Devices	PCs, laptops, smartphones, and web servers.	Sensors, smart actuators, smart bulbs, wearables, and edge nodes.
Communication Mode	Human-to-human or human-to-machine interaction.	Machine-to-machine (M2M) autonomous data exchange.
Data Types	User-generated text, images, video, and web files.	Telemetry, sensor readings, environmental metrics, and control signals.
Primary Examples	Sending emails, web browsing, streaming video.	Smart thermostats, automated irrigation, predictive factory maintenance.

Q7. Describe the 5-Layer IoT Architecture Model and detail the specific responsibilities of each layer.

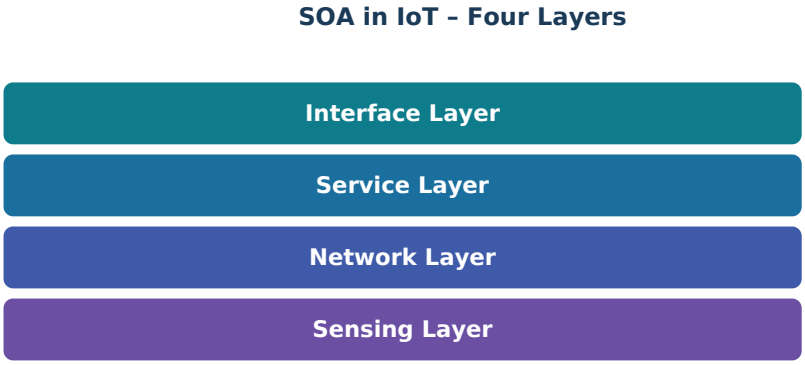
Answer:



1. **Perception Layer:** Represents the physical hardware layer housing sensors, actuators, and RFID tags that collect raw environmental data.
2. **Transport Layer:** Securely transfers collected sensor data from the perception layer to the processing layer via networks (Wi-Fi, 6LoWPAN, 5G).
3. **Processing Layer:** Stores, filters, and analyzes incoming data streams using cloud computing or database platforms.
4. **Application Layer:** Renders actionable information and control interfaces to end-users (smart home controls, health dashboards).
5. **Business Layer:** Manages overall IoT system operations, profit models, user privacy, and executive strategic planning based on application insights.

Q8. Explain Service-Oriented Architecture (SOA) in IoT, highlighting its four functional layers and key structural components.

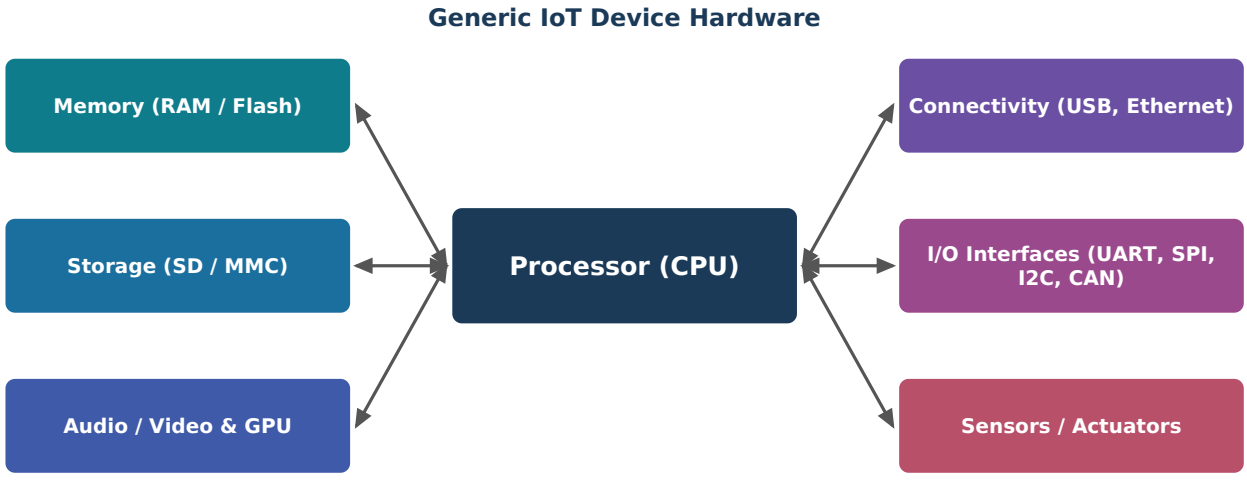
Answer:



1. **SOA Concept in IoT:** Encapsulates complex hardware functionality into abstract services published in a central registry for dynamic discovery and composition.
2. **Sensing Layer:** Hardware layer containing physical sensors and actuators that collect real-world physical data.
3. **Network Layer:** Transfers data from the sensing layer to the service layer using secure communication channels.
4. **Service Layer:** Creates, manages, and orchestrates software services requested by users and enterprise applications.
5. **Interface Layer:** Provides standardized APIs (RESTful endpoints, MQTT topics) allowing external clients to interact with managed services.

Q9. Describe the Generic Hardware Block Diagram of an IoT device, detailing the functions of the Processor, Connectivity, Memory, and I/O Interfaces.

Answer:



1. **Processor (CPU):** Executes firmware code, processes raw sensor inputs, and manages operating protocols (e.g., ARM Cortex).
2. **Connectivity Block:** Provides wired or wireless hardware network access via USB hosts, RJ45 Ethernet, or radio transceivers.
3. **Memory Block:** Combines permanent flash memory (NAND/NOR) for system firmware and DDR RAM for active runtime computation.

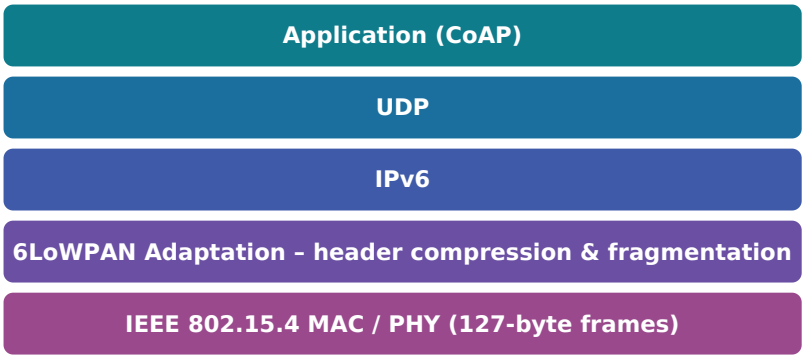
4. **I/O Interface Block:** Facilitates serial sensor connectivity using UART, SPI, I2C, and CAN bus buses.
5. **Storage & Graphics Interfaces:** Includes SD/MMC slots for local logging alongside optional GPU modules for visual displays.

Q10. Compare IPv4, IPv6, and 6LoWPAN at the network layer, explaining why 6LoWPAN is essential for constrained wireless networks.

Answer:

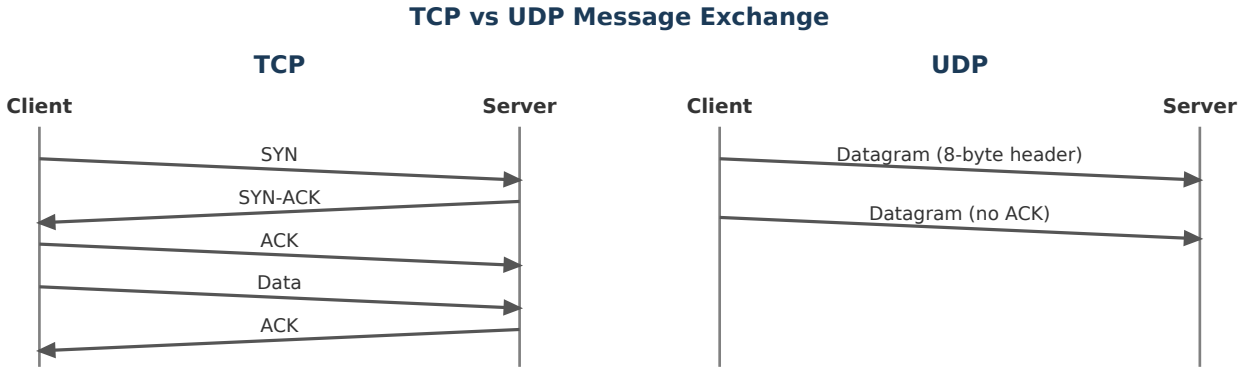
Feature / Protocol	IPv4	IPv6	6LoWPAN
Address Length	32-bit address space (~4.3 billion addresses).	128-bit address space (~3.4×10^38 addresses).	128-bit IPv6 address encapsulation.
Header Overhead	20 bytes (fixed base header).	40 bytes (fixed base header).	Compressed headers (down to a few bytes).
Target Devices	Legacy computers, routers, servers.	Modern servers, smartphones, gateways.	Resource-constrained low-power wireless sensors.
Adaptation Features	Requires NAT for address scaling.	Native autoconfiguration (SLAAC).	Built-in packet fragmentation & mesh routing.
Link Layer Binding	Ethernet (802.3), Wi-Fi (802.11).	High-bandwidth wired/wireless links.	IEEE 802.15.4 low-rate wireless networks.

6LoWPAN Protocol Stack



Q11. Analyze the trade-offs between TCP and UDP at the transport layer for constrained IoT environments.

Answer:

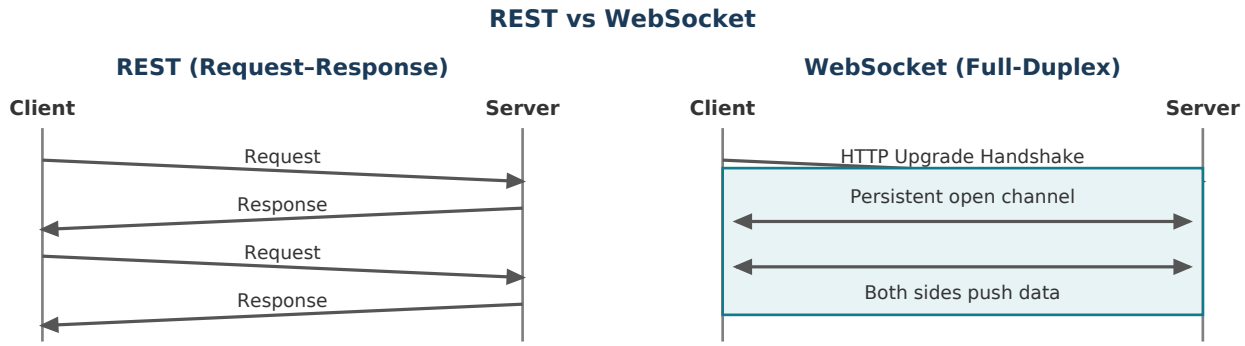


1. **TCP Connection Overhead:** TCP is connection-oriented, requiring a 3-way handshake and 20-byte header, which consumes energy on constrained sensors.
2. **TCP Reliability Mechanics:** Guarantees ordered, error-checked delivery using acknowledgments (ACKs) and retransmissions, making it ideal for firmware updates.
3. **UDP Lightweight Transmission:** UDP is connectionless and stateless with a tiny 8-byte header, transmitting datagrams with minimal processing latency.
4. **UDP Best-Effort Delivery:** Provides no inherent delivery guarantees, packet ordering, or flow control, transferring reliability responsibilities to application protocols like CoAP.
5. **IoT Application Binding:** TCP powers reliability-critical protocols (MQTT, HTTP), whereas UDP powers latency-sensitive streaming and low-power protocols (CoAP).

Q12. Compare REST-based APIs and WebSocket-based APIs in IoT communication architectures.

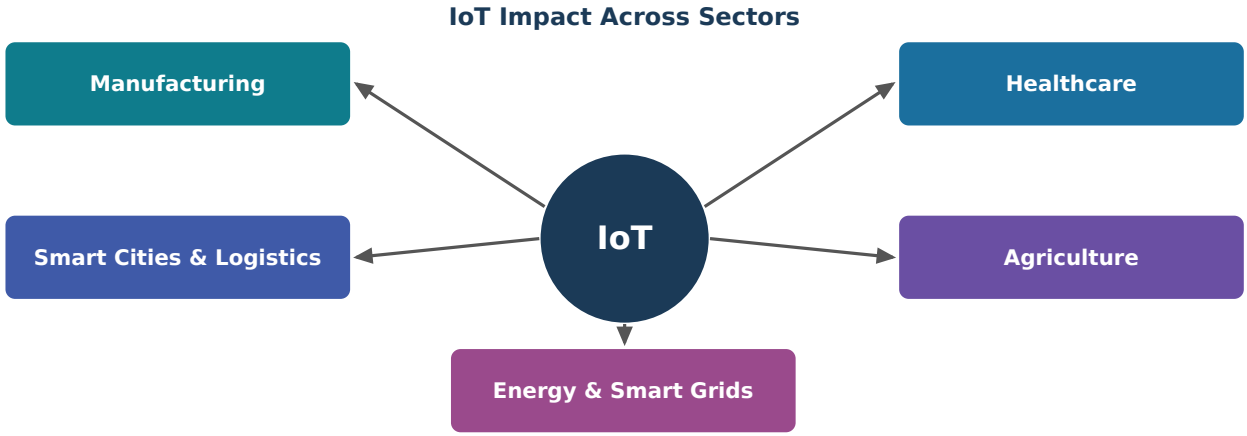
Answer:

Architectural Feature	REST-Based Communication APIs	WebSocket-Based Communication APIs
Communication Model	Request-Response model over HTTP.	Exclusive Pair, full-duplex continuous channel.
State Management	Strictly stateless; each request is independent.	Stateful; connection remains open after HTTP handshake.
Directionality	Unidirectional client-initiated request.	Bidirectional; both client and server push data freely.
Header Overhead	High overhead due to HTTP request/response headers.	Low overhead per frame after initial connection handshake.
Ideal IoT Use Cases	Periodic sensor queries, cloud database logging.	Real-time telemetry streaming, instant remote control.



Q13. Explain the key Economic and Industrial Impacts of IoT across manufacturing, healthcare, smart cities, agriculture, and logistics.

Answer:



1. **Smart Manufacturing (Industry 4.0):** Machinery vibration and temperature sensors enable predictive maintenance, preventing costly equipment breakdowns and reducing downtime.
2. **Healthcare & Remote Monitoring:** Wearable biosensors stream continuous patient vitals to clinicians, improving chronic disease management while reducing hospital visits.
3. **Smart Cities & Logistics:** Connected waste bins and smart street lights optimize municipal collection routes and energy usage, while GPS fleet tracking improves supply chains.
4. **Precision Agriculture:** Soil moisture sensors automate precision irrigation systems, conserving water and optimizing crop yields.
5. **Energy & Smart Grids:** Smart meters track grid voltage and load in real time, optimizing power distribution and integrating renewable energy sources efficiently.

Section C: 2-Mark Questions

Q14. Define the core concept of IoT and distinguish it from traditional machine communications.

Answer:

The Internet of Things is an interconnected network of physical objects embedded with sensing, processing, and communication hardware that autonomously exchange data over IP networks.

1. **Autonomous Operation:** Unlike traditional human-driven internet browsing, IoT operates via machine-to-machine (M2M) automation without manual human intervention.
2. **Physical World Integration:** It bridges digital cloud analytics with physical sensing and actuation in real time.

Q15. Explain the "Dynamic and Self-Adapting" characteristic of IoT devices with an example.

Answer:

Dynamic and self-adapting capability means an IoT system automatically adjusts its operating mode based on changing environmental conditions or user context.

1. **Automated Operational Shift:** Devices alter data sampling rates, power modes, or actions dynamically without manual reconfiguration.
2. **Real-World Example:** A smart thermostat adjusts air conditioning output based on ambient temperature, weather forecasts, and phone location detection.

Q16. What is a Unique Identity in IoT, and why is it required for network integration?

Answer:

Unique identity ensures that every physical device in an IoT network possesses a distinct, globally addressable identifier such as an IP address or URI.

1. **Individual Addressing:** Allows individual devices among billions to be distinctly identified, queried, and managed over networks.
2. **Hardware Binding:** Typically bound to physical hardware identifiers like MAC addresses or cryptographic device certificates.

Q17. Define Edge Computing and state its main advantage in the IoT World Forum model.

Answer:

Edge computing performs data filtering, aggregation, and analytics locally on hardware nodes situated near the physical sensors.



1. **Bandwidth Optimization:** Filters raw sensor noise locally so that only critical alerts or summaries are sent to cloud servers.
2. **Latency Reduction:** Enables sub-millisecond local automated responses for safety-critical industrial actions.

Q18. What is 6LoWPAN, and what problem does header compression solve in low-power networks?

Answer:

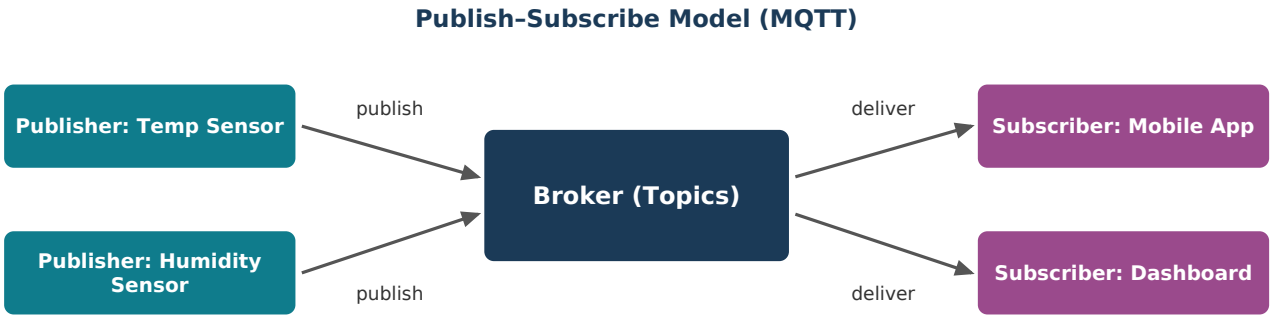
6LoWPAN enables IPv6 communication over low-power, low-data-rate IEEE 802.15.4 wireless personal area networks.

1. **Frame Size Limitation:** IEEE 802.15.4 frames carry a maximum payload of 127 bytes, which cannot fit standard 40-byte IPv6 headers.
2. **Header Compression Solution:** 6LoWPAN compresses IPv6/UDP headers down to a few bytes, allowing native IP routing without wasting battery or bandwidth.

Q19. Briefly define the Publisher-Subscriber communication model in IoT.

Answer:

The Publisher-Subscriber model is an asynchronous messaging pattern where message senders (publishers) and receivers (subscribers) are decoupled via a central broker.

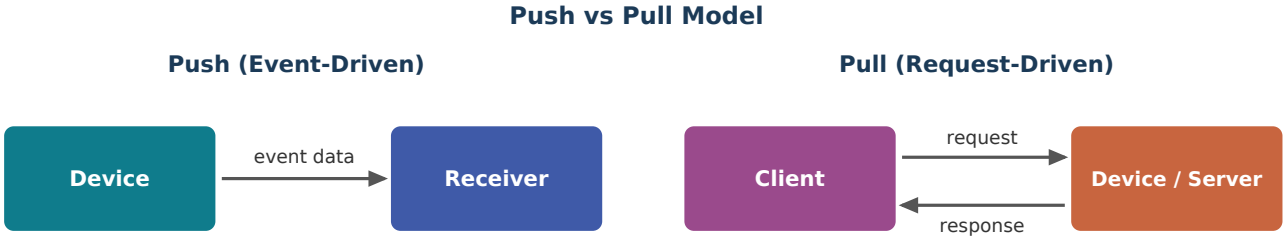


1. **Topic-Based Routing:** Publishers send data packets tagged with specific topic strings to the broker without knowing who the receivers are.
2. **Broker Distribution:** The central broker automatically forwards received messages to all clients currently subscribed to that topic.

Q20. Differentiate between the Push Model and Pull Model in IoT data exchange.

Answer:

Push and Pull models define whether data transmission is initiated automatically by the sensing device or requested on demand by the client.



1. **Push Model (Event-Driven):** The device automatically pushes data to receivers whenever an event triggers or a timer expires (e.g., motion sensor alert).
2. **Pull Model (Request-Driven):** The client explicitly requests data from the device or server, which responds on demand (e.g., fetching weather status on dashboard open).

Q21. List two major Security and Privacy challenges in IoT deployments.

Answer:

1. **Vulnerable Resource-Constrained Hardware:** Tiny sensors lack CPU capacity for standard heavy encryption, leaving them susceptible to unauthorized access.
2. **Pervasive Data Privacy Risks:** Extensive continuous data collection from personal or industrial spaces risks unauthorized data exposure.

Q22. What are the primary network drivers requiring Ultra-Low Latency in modern IoT architectures?

Answer:

1. **Mission-Critical Applications:** Driven by autonomous vehicle navigation, industrial robotics, remote telesurgery, and smart grid tripping mechanisms.
2. **Edge Architecture Shift:** Requires shifting compute workloads from distant cloud data centers to local 5G edge computing nodes.